Week 7 — While Loops & The Guessing Game

Name: ____ Date: ____

> *A `while` loop repeats **until its condition goes False**. No update line → it never stops!*

Today's words

Word	What it means
`while` loop	repeats as long as its condition stays True
condition	the True/False check before every lap
update line	the line inside that changes the variable — the escape route
infinite loop	a loop that never stops (escape hatch: **Ctrl+C**)
binary search	always guess the middle — half the suspects vanish each time

1 · Be the computer: trace the loop 🤖

```
n = 6
while n > 0:
    print(n)
    n = n - 2
print("done")
```

lap	what prints	`n` after the update
1		
2		
3		

Last line on the screen: ______

Danger question: if the line `n = n - 2` were deleted, what would happen?

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2 · The halving chain ⛓️

Each higher/lower answer cuts the suspects in half (round down). Fill in the chain:

100 → 50 → 25 → _ → _ → ____ → 1

Count the numbers in the chain: _ ***— that's why*** _ guesses always win. Check: 2^7 = ____, which is bigger than 100. ✔

3 · Binary search on paper 🔎

A friend picks **11**, from 1 to 16. You always guess the **middle** of what's left (round down). Finish the hunt — row 1 is done for you:

numbers still possible	your guess	friend says
1–16	8	higher!
9–16		
		correct! 🎉

🧠 Brain teaser (optional — take it home)

Seven guesses crack 1–100 because 2^7 = 128 > 100. **How many guesses would you need for 1 to 1,000? For 1 to 1,000,000?** (Hint: keep doubling — 2, 4, 8, 16... — and count how many doublings pass each target.) Then the mind-bender: going from 1,000 to 1,000,000 multiplies the suspects by a THOUSAND — how many *extra* guesses did it cost? Bring your answer next week for a shout-out.